

Docker Container Memory Leak SOP

Objective

Identify and mitigate increasing memory usage in Docker containers that may lead to OOM exits, degraded performance, or host instability.

Scope

Applicable to Docker-based services, containerized application deployments, DevOps teams, SRE teams, platform teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

1. Access to Docker host, Docker CLI, container runtime logs, and the target container or image name.
2. Permission to inspect containers, images, volumes, networks, resource limits, and compose files where applicable.
3. Access to application logs, deployment artifacts, Dockerfile, environment variables, and registry credentials.
4. Monitoring access for host CPU, memory, disk, container restart count, and container health checks.
5. Escalation contact for application, DevOps, security, registry, infrastructure, and platform teams.

Incident Metadata

Field	Value
Ticket ID	T032
Issue	Docker container memory leak
Category	Performance
Service	Docker
Severity	High
Status	Resolved
Description	Container memory usage keeps increasing
Expected Resolution	Fix memory leaks in application code

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
L001	5/1/2026 10:00	ERROR	Container exited unexpectedly	DOCKER_137
L006	5/1/2026 10:03	WARNING	High memory usage detected	DOCKER_MEM
L026	5/1/2026 10:13	ERROR	High memory usage detected	DOCKER_MEM
L042	5/1/2026 10:21	ERROR	Container exited unexpectedly	DOCKER_137
L046	5/1/2026 10:23	WARNING	High memory usage detected	DOCKER_MEM

Procedure

1. Confirm container name, memory trend, OOM event, exit code, and impacted service.
2. Use docker stats and monitoring dashboards to compare current and historical memory usage.
3. Review logs for large payloads, cache growth, unbounded queues, or repeated failures.
4. Inspect container memory limits, swap settings, and host memory pressure.
5. Check recent code, dependency, configuration, or traffic changes.
6. Profile the application or capture heap dump where supported.
7. Apply mitigation such as restart, scale out, memory limit adjustment, cache cleanup, or code fix.

8. Monitor memory usage after redeployment and confirm no steady increase remains.

Common Commands

```
docker ps -a
docker logs <container-name-or-id> --tail=100
docker inspect <container-name-or-id>
docker stats --no-stream
docker images
docker network ls && docker network inspect <network-name>
docker volume ls && docker volume inspect <volume-name>
docker system df
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Memory keeps increasing	Application memory leak	Profile and fix leak
Container exits with 137	OOM kill	Increase limit or reduce memory usage
Memory spike on workload	Large batch or payload	Chunk processing or scale service
Host memory pressure	Too many high-memory containers	Move workload or add capacity

Best Practices

1. Set memory limits based on real usage data.
2. Alert before memory reaches critical threshold.
3. Avoid unbounded in-memory caches.
4. Capture memory diagnostics for repeated incidents.

References

Docker Docs | Internal Tickets CSV | Internal Logs CSV | Container Runtime Logs | Monitoring Dashboards | DevOps Runbook